Properties of spin-lattice relaxation of Yb^{3+} ions in phosphate glasses

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The results of experimental and theoretical investigations of the temperature, concentration, and size dependences of the spin-lattice relaxation rate of Yb^{3+} ions in phosphate glasses by the electron spin echo method are presented. A model is proposed for the spin-lattice relaxation that makes it possible to explain the experimental results if the specific properties of the glass state are taken into account. © *1995 American Institute of Physics.*

In inorganic glasses the spectral density of the vibrational states at low energies has two features which distinguish it from a Debye spectrum. At energies below 1 K the vibrational density of states is constant and in the energy range 3-15 K it is two to six times higher than the Debye density of states. The first feature is explained by the standard model of two-level centers, but the nature of the second is not completely understood. In the models which have been developed,[1] however, it is assumed that this feature is due to vibrational excitations, localized on scales of 1–2 nm and carrying information about the structure of the glasses. It can be conjectured that localized vibrational excitations should be manifested in spin-lattice relaxation (SLR) of impurity paramagnetic ions in glasses.[2,3] However, this has not been determined in experimental investigations of SLR.[4–6]

In the present paper we present the results of an investigation of SLR of Yb^{3+} ions in phosphate glass (PG). We note that this object has been studied previously by the electron spin echo (ESE) method,[7] but the main objective of Ref. 7 was to demonstrate that the ESE method provides useful information in the investigation of glasses. No significant peculiarities of SLR inherent to glass were observed. Our objective in the present work was to determine the possible anomalies of the direct and Raman SLR processes which are determined by the structural features of the glass and to establish the effect of localized vibrations on SLR.

The experiments were performed by the ESE method, which is more effective for measuring the SLR times in activated glasses than is the traditional stationary EPR method of pulsed saturation of transitions. The main reason for this is that the ESE method makes possible a spectrally selective study of the spin packets of inhomogeneously broadened EPR lines of rare-earth elements in glasses and to investigate their dynamics.

Phosphate glasses with the composition $75P_2O_5 \cdot 25La_2O_3$ were investigated. The ytterbium content was varied by substituting Yb_2O_3 for La_2O_3 with the following concentrations x(mol%): 0.03 (I), 0.1 (II), and 0.3 (III). The samples were synthesized at the All-Union Scientific Center "Vavilov State Optics Institute" in St. Petersburg.

The rare-earth Yb^{3+} ions are a convenient paramagnetic probe for studying direct and Raman SLR processes. The Orbach SLR processes are usually inefficient, since the energy of the first excited doublet of the Yb^{3+} ion is, as a rule, quite high. The unfilled shell of the Yb^{3+} ion has a $4f^{13}$ electronic configuration, and the electric field of the matrix splits the $F^2_{7/2}$ ground state into four Kramers doublets. The structure of the Stark levels of the ground term of the Yb^{3+} ion in phosphate glass was proposed in Ref. 8 on the basis of optical spectra. The energies of the excited levels of the lowest term are ~238, 464, and 780 cm^{-1}. In Ref. 8 it was concluded that the octahedral environment of the rare-earth ion is distorted. The results of the investigation of the EPR spectra by the ESE methods[9] agree with this conclusion.

1. EXPERIMENTAL RESULTS

The spin-lattice relaxation times T_1 were measured on a ESE spectrometer operating at a frequency of 9.4 GHz with a time resolution of 10^{-7} sec in the temperature range 1.6-7 K. To prevent heating of the sample, some measurements were performed in liquid helium.

The EPR spectrum of Yb^{3+} ions, measured with a two-pulse ESE, is shown in Fig. 1. The plot displays the EPR spectra normalized to the maximum amplitude of the echo. The fact that the EPR line belongs to the ytterbium ions is proved by the fact that the line intensity is proportional to the amount of ytterbium oxide introduced into the charge. The width of the EPR line of Yb^{3+} is comparable to the magnitude of the magnetic field corresponding to the maximum intensity of the EPR line.

The SLR time was measured using the well-known method of Ref. 10, in which a long microwave pulse, which saturates the spin system, is introduced before the pair of echo-forming pulses. The pulse duration was set at a value so that the exponential tail of the magnetization restoration curve did not change with a further increase of the power of the saturating pulse. For saturating-pulse powers $P \sim 50$ mW and 1 W, the pulse duration was equal to 2 msec and 200 μsec, respectively.

The experiments showed that the kinetics of the restoration of the ESE pulse after the saturating pulse was nonexponential, and the SLR times were extracted from the exponential tail. However, in the temperature range where Raman processes predominate, the relaxation curve was virtually exponential. In the temperature range where direct SLR processes predominate, the nonexponentiality of the kinetics was even greater. Figure 2 displays the typical kinetics of the recovery of the ESE amplitude after the saturating pulse. The

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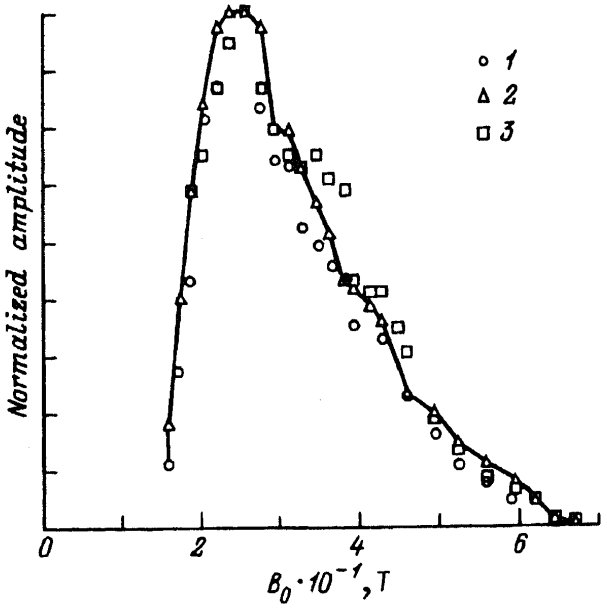


FIG. 1. EPR spectrum of Yb^{3+} ions in phosphate glass. $T=4.2$ K; $\nu=9.4$ GHz; x (mole %): 1—0.03 (sample I), 2—0.1 (sample II), 3—(sample III).

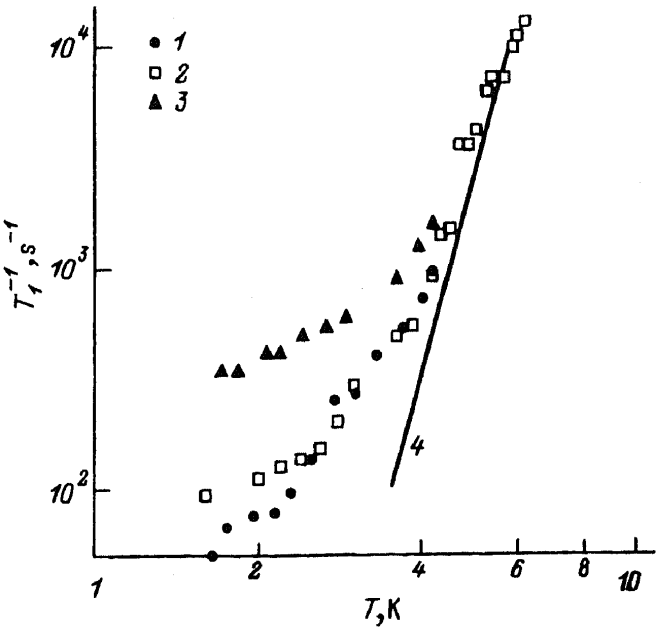


FIG. 3. Temperature dependences of the spin-lattice relaxation rates of Yb^{3+} ions in phosphate glass. $\nu=9.4$ GHz; x (mole %) and $B_0(T)$, respectively, are: 1—0.03 (sample I) and 0.5, 2—0.03 (sample I) and 0.26, 3—0.3 (sample III) and 0.5, 4—the temperature dependence $1/T_1 \sim T^9$.

temperature dependences of the SLR rates were measured for two values of the external magnetic field: at the maximum of the EPR line at $B_0=0.26$ T, and on the strong-field wing at $B_0=0.5$ T. The experimental temperature dependences of the SLR rate T_1^{-1}, the concentrations, and the size of the samples are presented in Figs. 3–5. We note the following behavior of the results obtained.

The temperature dependences of the SLR rate T_1^{-1} at $T>4$ K for all samples investigated are described (within the limits of error of the experiment) by the Raman process $T_1^{-1} \sim T^9$. Figure 3 displays the temperature dependences of T_1^{-1} for sample I with the lowest concentration of Yb^{3+} atoms for two values of the magnetic field and for sample III with the highest concentration of the impurity ions in a field $B_0=0.5$ T. The straight line in Fig. 3 represents the temperature dependence $T_1^{-1} \sim T^9$. At temperatures $T<4.0$ K it was observed that T_1^{-1} depends on the concentration of the activator and on the magnetic field.

The dependence of T_1^{-1} on the size of the sample was investigated in the temperature range where the direct process predominates. For this, the samples I and III of phosphate glass were cut out in the form of a $3\times2.6\times2.2$ mm rectangular parallelepiped. Next, one face of the sample, perpendicular to the smallest dimension l, was systematically ground down. The size dependence $T_1^{-1}(l)$ appears for $l<0.5$ mm. The results of the measurements are presented in Fig. 4.

For samples II and III with high Yb^{3+} concentration a plateau — a temperature-independent section — is observed in the temperature dependence of T_1^{-1} in a field $B_0=0.26$ T and in a narrow range of fields (several hundredths of a tesla) (Fig. 5). Moreover, the rate T_1^{-1} is higher in the sample with

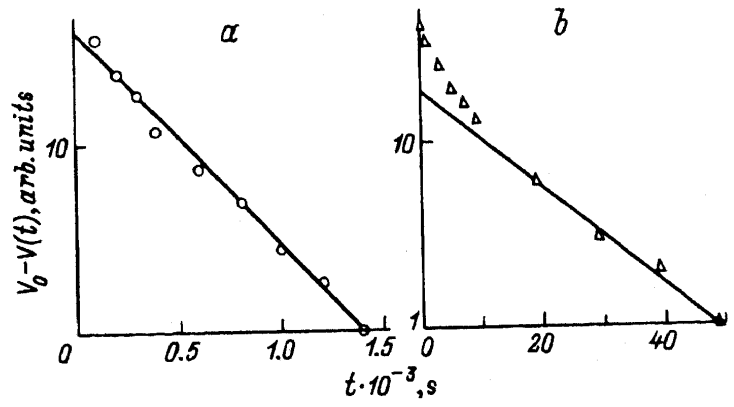


FIG. 2. Kinetics of the recovery of the ESE amplitude $V_0-V(t)$ after the saturating pulse. The straight line represents the function $V_0-V(t)=V_0\exp(-t/T_1)$. $x=0.03$ mole % (sample I); T (K): a—3.9, b—2.0.

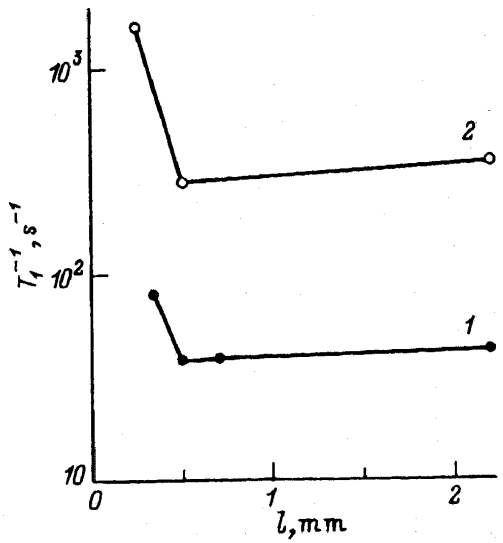


FIG. 4. Spin-lattice relaxation rate of Yb^{3+} ions in phosphate glass as a function of the sample size l. $T=1.65$ K; $B_0\approx0.5$ T; $\nu=9.4$ GHz; x (mole %): 1—0.03 (sample I), 2—0.3 (sample III).

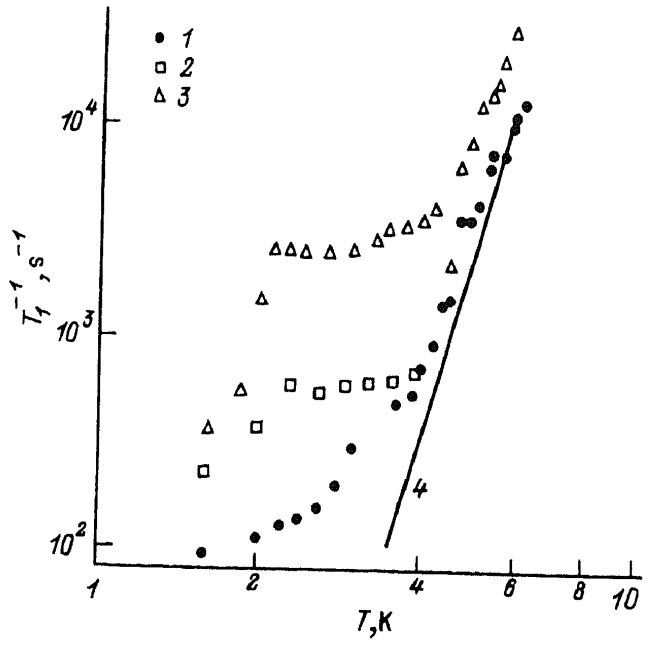


FIG. 5. Temperature dependences of the spin-lattice relaxation rate of Yb^{3+} ions in phosphate glass. $B_0 \approx 0.26$ T; $\nu = 9.4$ GHz; x (mole %): 1—0.03 (sample I), 2—0.1 (sample II), 3—0.3 (sample III), 4—the temperature dependence $1/T_1 \sim T^9$.

a high activator concentration x. The results obtained, taken together, suggest that resonance cross relaxation is observed at low temperatures.

Taking into consideration what we have said above, the temperature dependence of the rate T_1^{-1} at $B_0 = 0.26$ T was interpreted as a sum of three contributions

$$T_1^{-1} = AT + CT^9 + W(x), \tag{1}$$

where the first term is the direct process, the second term is the Raman process, and the third term describes the cross-relaxation process, which depends on the concentration of Yb^{3+} ions. The cross-relaxation process ensures transfer of spin excitation to a rapidly relaxing center (RRC) that relaxes directly into the lattice. The nature of the RRC and the functional form of $W(x)$ are discussed below.

2. DISCUSSION OF THE RESULTS

a) The effect of the localized vibrations, which are characteristic of glasses, on the SLR of a paramagnetic impurity can be manifested in the fact that the Debye temperature Θ_{slr}, extracted from the description of the experimental temperature dependences of the SLR rate of the Raman process, is found to be much lower than the Debye temperature Θ_d obtained by standard methods. An analogous behavior of Θ_{slr} was observed in investigations of SLR in crystals.[11–14] In Refs. 11 and 15 it was shown that the experimentally observed relation $\Theta_{slr} < \Theta_d$ is associated with the fact that the real phonon spectrum is different from the Debye spectrum. In addition, Θ_{slr} was found to be different for different ions in the same matrix and for the same ions occupying different sites of the crystal lattice. This suggested that Θ_{slr} characterizes a definite part of the real phonon spectrum of a crystal that is responsible for the interaction with the paramagnetic ions.

The fact that in glasses the vibrational spectral density is different from the Debye density is well known, and the peculiarity of the vibrational spectrum at temperatures in the range 3–15 K is associated with localized vibrations. For phosphate glasses, the Debye temperature Θ_d can be calculated using the formula of Ref. 16 and the measured values of the sound speeds in the phosphate glass:[17] $v_l = 4.57 \cdot 10^3$ m/sec (longitudinal sound) and $v_t = 2.68 \cdot 10^3$ m/sec (transverse sound). The average sound speed v_m can be calculated from the formula $v_m = [1/3(2/v_t^3 + 1/v_l^3)]^{-1/3} = 2.97 \cdot 10^3$ m/sec. The result is $\Theta_d = 350$ K. The value of Θ_{slr} for Yb^{3+} in phosphate glass could not be determined by the ESE method because the phase relaxation times are too short. However, the value $\Theta_{slr} = 160$ K for Yb^{3+}-activated phosphate glass is known from investigations of the temperature dependence of the SLR rate T_{1N}^{-1} of ^{31}P nuclei.[17] The temperature dependence of T_{1N}^{-1} is determined by the paramagnetic relaxation of Yb^{3+} ions and has the form $T_{1N}^{-1} \sim T^9 (160/T)$. Therefore, for Yb^{3+}-activated phosphate glass $\Theta_{slr} < \Theta_d$. We conjecture that this inequality is due to the localized vibrations which are characteristic of glass.

An example of the relation $\Theta_{slr} < \Theta_d$ for glasses is also given in Ref. 4. In describing the experimental results for ytterbium-activated silicate glass using the relation $T_1^{-1} \sim CT^9$ the value $\Theta_{slr} = 40$ K was obtained. From different estimates and measurements, it is known that the Debye temperature for this glass ranges from 312 to 475 K. This suggests that a significant part of the vibrational spectrum of the lattice in the glass studied in Ref. 4 refers to localized excitations.

b) We shall now analyze the contribution of cross-relaxation to SLR, represented by $W(x)$ in Eq. (1) and associated with RRC. The experimental results were analyzed by the method described in Ref. 18 for the case of a uniform distribution of impurities. The model of SLR through RRC, as a result of cross-relaxation, is displayed in Fig. 6. The solution of the kinetic equations for the system under study leads to two exponentials in the restoration curve of EPR signal with characteristic relaxation times τ_1 and τ_2. For $N_1 \gg N_2$ (Ref. 18)

$$\tau_1^{-1} \approx \frac{1}{T_1} + \frac{\dfrac{N_2}{N_1 T_{21}}\left(\dfrac{1}{T_1^*} - \dfrac{1}{T_1}\right)}{\dfrac{1}{T_{21}} + \dfrac{1}{T_1^*} - \dfrac{1}{T_1}}, \tag{2}$$

$$\tau_2^{-1} \approx \frac{1}{T_1^*} + \frac{1}{T_{21}}. \tag{3}$$

In Eqs. (2) and (3) N_1 is the number of the main Yb^{3+} ions ($N_1 = x$) and N_2 is the number of RRCs; T_1 and T_1^* are the SLR times of Yb^{3+} and the RRC, respectively; and, T_{21} is the cross-relaxation time. The equations (2) and (3) lead to very diverse temperature and concentration dependences of the SLR rates.

In the particular case (i) of fast SLR of RRCs, when $1/T_1^* \gg 1/T_{21} \gg N_1/N_2 T_1$, we have from Eqs. (2) and (3) $\tau_1^{-1} \sim N_2/N_1 T_{21}$ and $\tau_2^{-1} \sim 1/T_1^*$. For this situation sections where the SLR rate is temperature independent are charac-

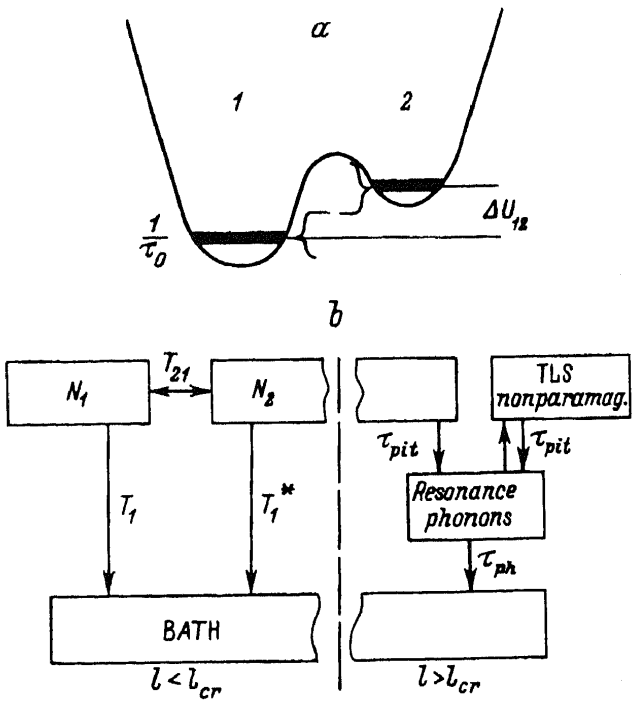


FIG. 6. Scheme of spin-lattice relaxation of Yb^{3+} ions via RRC as a result of cross-relaxation.

teristic. However, even for the opposite limit (ii) of fast cross relaxation $1/T_{21} \gg 1/T_1^* \gg 1/T_1$, we have from Eqs. (2) and (3) $\tau_1^{-1} \sim N_2/N_1 T_1^*$ and $\tau_2^{-1} \sim 1/T_{21}$. Here it is possible to have a situation when the SLR rate is temperature independent. We shall consider the possible nature of these SLR processes and the proposed model in greater detail.

For RRCs we shall consider a local paramagnetic center which has a multiwell potential and which tunnels between states of the different wells. We shall discuss below the case of states localized in separate wells; this corresponds to a tunneling matrix element which is small compared to the natural line width $1/\tau_0$ or compared to the splitting ΔU_{ij} between the single-well states in the internal random (defect) field.

In Refs. 19–22 it was shown that in such a situation a RRC appears when an external magnetic field B_0 is applied, and this is when the average Zeeman splitting $\beta B_0(g_1+g_2)/2$, where g_1 and g_2 are the g-factors of the spin states in the potential wells 1 and 2, between which the tunneling transition occurs, is equal to (to within $1/\tau_0$) the corresponding value of ΔU_{12}. In this case the states with different spin projection and in different wells are found to be in resonance. If the spin-flip tunneling transitions (primarily as a result of spin-orbit tunneling[19]) between the states is taken into consideration, then a RRC arises. The SLR rate of such a RRC in the situation of a bottleneck on tunneling transitions is represented in the form[19–22]

$$\frac{1}{T_1^*} = \frac{|\lambda'|^2}{\frac{1}{\tau_0} + \left(\Delta U - \frac{g_1+g_2}{2}\beta B_0\right)^2 \tau_0}. \tag{4}$$

Here λ' is the tunneling matrix element for a transition with a change of the spin state. The quantity $1/\tau_0$ in the situation of the paramagnetic system consists of two contributions[22]

$$\frac{1}{\tau_0} = \frac{1}{\tau_{ph}} + \frac{1}{\tau_{ss}}, \tag{5}$$

where $1/\tau_{ph}$ is related with the elastic scattering of phonons by single-well states (which, in particular, leads to $1/\tau_{ph} \sim T^7$) and $1/\tau_{ss}$ is associated with the spin-spin cross-relaxation. In the situation of high concentration of paramagnetic centers (when $1/\tau_{ss}$ is quite high) and low temperatures (when $1/\tau_{ph}$ is quite small), considered here, we have

$$\frac{1}{\tau_{ss}} \gg \frac{1}{\tau_{ph}}, \tag{6}$$

which in the resonance situation $\Delta U \approx (g_1+g_2)B_0\beta/2$ leads to

$$\frac{1}{T_1^*} \cong |\lambda'|^2 \tau_{ss}. \tag{7}$$

It follows from Eq. (7) that $1/\tau_1^*$ exhibits a weak temperature dependence and decreases with increasing concentration of the paramagnetic centers.

For further analysis, two possible cases of tunneling RRCs in glass must be distinguished. The first case is when the RRC is formed by a paramagnetic tunneling center which is not associated with the paramagnetic Yb^{3+} impurity (for example, broken bonds). In the second case the RRC consists of a Yb^{3+} paramagnetic impurity located near a two-level tunneling system in glass.

If $\tilde{g}(\Delta U)$ is the distribution function of the energy splittings of the one-well states in glass, then in the first case the number of RRCs depends on the concentration and the magnetic field

$$N_2 \sim \frac{1}{\tau_{ss}(x)} \tilde{g}\left(\frac{(g_1+g_2)}{2}\beta B_0\right), \tag{8}$$

while in the second case

$$N_2 \sim \frac{x}{\tau_{ss}(x)} \tilde{g}\left(\frac{(g_1+g_2)}{2}\beta B_0\right). \tag{9}$$

As a result, in the first case for the limit (i), $1/T_1^* \gg 1/T_{21}$, as follows from Eqs. (2)–(4) and (7), we have

$$\frac{1}{\tau_1} \sim \frac{1}{x}\left(\frac{1}{\tau_{ss}(x)}\right)^2 \tilde{g}\left(\frac{(g_1+g_2)}{2}\beta B_0\right), \tag{10}$$

$$\frac{1}{\tau_2} \approx |\lambda'|^2 \tau_{ss}(x), \tag{11}$$

while for the second case

$$\frac{1}{\tau_1} \sim \left(\frac{1}{\tau_{ss}(x)}\right)^2 \tilde{g}\left(\frac{(g_1+g_2)}{2}\beta B_0\right), \tag{12}$$

$$\frac{1}{\tau_2} \approx |\lambda'|^2 \tau_{ss}(x). \tag{13}$$

In the limit of fast spin-spin cross-relaxation (ii), $1/T_{21} \gg 1/T_1^*$, we obtain different dependences. For example, in the first case, as follows from Eqs. (2)–(4) and (7), we have

$$\frac{1}{\tau_1} \sim \frac{|\lambda'|^2}{x}\tilde{g}\left(\frac{(g_1+g_2)}{2}\beta B_0\right), \tag{14}$$

$$\frac{1}{\tau_2} \approx \frac{1}{\tau_{ss}(x)}, \tag{15}$$

and in the second case

$$\frac{1}{\tau_1} \sim |\lambda'|^2\tilde{g}\left(\frac{(g_1+g_2)}{2}\beta B_0\right),$$

$$\frac{1}{\tau_2} \approx \frac{1}{\tau_{ss}(x)}. \tag{17}$$

In what follows, we shall consider a Gaussian distribution $\tilde{g}(\Delta U)$. This corresponds to, specifically, the case of strong correlation between tunneling centers, but in a situation when the temperature of the sample is higher than the transition temperature T_g into the phase of a pseudospin orientational glass. The features of $\tilde{g}(\Delta U)$ in the temperature range $T<T_g$ and the corresponding change of the SLR will be discussed later. But, at least for $T \geqslant T_g$ (where $T_g \cong 1.5$ K, as will be shown below) the use of a Gaussian as a model distribution function over ΔU in the case at hand appears to be justified. As a result we obtain

$$\tilde{g}\left(\frac{(g_1+g_2)}{2}\beta B_0\right) = \frac{1}{\sqrt{2\pi\sigma}}\exp - \left(\frac{\left(\frac{g_1+g_2}{2}\beta B_0\right)^2}{2\sigma}\right), \tag{18}$$

where the second moment σ of the distribution function can be represented, in our case, in the form $\sigma=0.5(\alpha+\beta x)$. The last term in this expression describes the broadening induced by the Yb^{3+} impurity.

The main features of the behavior of the cross-relaxation part of the SLR which we shall use for the analysis are as follows: 1) the existence of temperature-independent sections in $1/T_1(T)$; 2) the quite large value of $1/T_1$ on these sections; 3) appreciable acceleration of SLR with increasing concentration; and, 4) substantial slowing of SLR in this region at the transition to the high-field wing of the EPR line, where the magnetic field is approximately two times stronger than at the center of the line (the observed similar behavior of SLR at the transition to the low-field wing of the EPR line can be associated to contributions from other centers and will not be analyzed on the basis of the simple model proposed here). On the basis of these experimental facts and the preceding analysis, we conclude that although the existence of the temperature-independent acceleration of SLR and retardation of SLR with increasing magnetic field can be explained in all of the four limits considered above, the concentration dependence of $1/\tau_1$ makes it possible to leave for further matching and comparison with experiment only two of them, namely, the first case in the limit (i) of strong RRC, when

$$\frac{1}{\tau_1} \sim \frac{x}{\sqrt{\alpha+\beta x}}\exp - \left(\frac{\left(\frac{g_1+g_2}{2}\beta B_0\right)^2}{\alpha+\beta x}\right), \tag{19}$$

$$\frac{1}{\tau_2} \sim \frac{1}{x}, \tag{20}$$

and the second case in the limit (ii) of fast spin-spin cross-relaxation

$$\frac{1}{\tau_1} \sim \frac{1}{\sqrt{\alpha+\beta x}}\exp - \left(\frac{\left(\frac{g_1+g_2}{2}\beta B_0\right)^2}{\alpha+\beta x}\right), \tag{21}$$

$$\frac{1}{\tau_2} \sim x. \tag{22}$$

Here, we took into account the fact that $1/\tau_{ss} \propto x$. The experimental results from the investigation of $1/\tau_1(x,B_0)$ dependences can be explained on the basis of the expression (19) with $\alpha=(1.1\pm 0.3)\cdot 10^{-47}$ and $\beta=(3.7\pm 0.5)\cdot 10^{-66}$ and on the basis of the expression (21) with $\alpha=(7.0\pm 0.3)\cdot 10^{-48}$ and $\beta=(2.1\pm 0.5)\cdot 10^{-66}$. A choice between these two situations can be made by comparing the experimental behavior of $1/\tau_2(x)$ and the model limiting expressions (20) and (22) for $1/\tau_2(x)$, which indicate that in these two situations $1/\tau_2(x)$ exhibits the opposite behavior. The experiment showed that $1/\tau_2(x)$ definitely increases with x, and according to a law corresponding to $1/\tau_2(x) \sim x$ within the limits of the experimental error (for $x=0.3$ mol%, $1/\tau_2(x)=6.45\cdot 10^3$ s^{-1}, while for $x=0.1$ mol% $1/\tau_2(x)=1.9\cdot 10^3$ s^{-1}). This indicates that the limit (ii) of fast spin-spin relaxation, when the bottleneck of the entire SLR process is determined by the RRC, is realized. The RRCs are Yb^{3+} ions located near the two-level tunneling systems in the glass.

The present model also makes it possible to explain the sharply lower SLR rate at $T \leqslant 1.5$ K. This decrease could be connected with the fact that the two-level tunneling systems (whose concentration can be much higher than the concentration of the Yb^{3+} paramagnetic centers) effectively interact with one another as a result of the indirect interaction via acoustic phonons.[23,24] For this reason, as the temperature decreases there can occur a percolation phase transition, as a result of which the system of tunneling pseudospins ends up in an orientational-glass phase. Calculations show that for a reasonable concentration of two-level tunneling systems of the order of $x \sim 0.1\%$ and equilibrium distortions of the order of ~ 0.5 Å at their location, the temperature T_g of the transition into the orientational pseudospin glass phase is found to be of the order of several degrees Kelvin. This agrees very well with the assumption that in the case at hand $T_g \cong 1.5$ K. It is this assumption that makes it possible to explain the sharply lower SLR rate at $T \leqslant 1.5$ K.

Indeed, in the orientational pseudospin glass phase the distribution function $\tilde{g}(\Delta U)$ has a sharp dip and a minimum at $\Delta U \approx 0$. This is associated with the appearance of short-range order in the pseudospin system as a result of correlations between their orientations. As a result, because of their relative smallness the values of $\Delta U=(g_1+g_2)/2\beta B_0$ which

are important for the RRC effect for the magnetic fields of the order of 0.5 T lie in the region of this dip in $\tilde{g}(\Delta U)$. This greatly diminishes the number of RRCs at temperatures $T \leqslant T_g = 1.5$ K and thereby also the SLR rate at temperatures $T \leqslant 1.5$ K).

c) An unexpected size effect was observed at temperatures corresponding to the range where the direct SLR process predominates (Fig. 4). It is natural to conjecture that the observed critical size $l \approx 0.5$ mm, which is a threshold for manifestation of the size effect, is associated with the characteristic times over which equilibrium is established in the vibrational system of the glass. On the other hand, three vibrational subsystems which participate in the process of low-temperature SLR must be distinguished in the glass. These are: (j) the temperature bath of thermal phonons, (jj) the system of resonant phonons and, (jjj) two-level tunneling systems, whose spectra contain energy gaps, resonant with the EPR photon and appearing as RRC (i.e., there are no Yb^{3+} ions near them). In this connection, we shall examine in greater detail the process of SLR of RRC which predominates in our case.

The RRC spin-phonon relaxation process itself consists of two successive processes: resonant spin-flip tunneling (with a rate $1/T_1^*$) and phonon-induced tunneling (PIT) at the frequency of the EPR photon, which occurs with conservation of the spin state. The rate $1/\tau_{pit}$ of the latter process determines not only the rate of production of resonant phonons but also the rate of absorption of a resonant phonon by a two-level tunneling system (not associated with a Yb^{3+} ion) and the subsequent reemission of the phonon by the system, as a result of which the two-level tunneling systems participate in the trapping of resonant phonons. Finally, the third important parameter of the problem is the rate $1/\tau_{ph}$ at which thermodynamic equilibrium is established for long-wavelength phonons. For the low temperatures considered here, this rate is determined by anharmonic processes and scattering by the surface of the sample

$$\frac{1}{\tau_{ph}} = \frac{1}{\tau_{ph}^{(1)}} + \frac{1}{\tau_{ph}^{(a)}}. \tag{23}$$

The first term in Eq. (23) can be represented in the form $1/\tau_{ph}^{(1)} \approx (l/v)^{-1}$, where v is the sound speed and l is the length of the sample. An important factor is that resonant spin-flip tunneling is a bottleneck in the SLR of RRCs. This follows directly from the character of the temperature dependence of SLR. This means that $1/\tau_{pit} > 1/T_1^*$ and $1/\tau_{ph} > 1/T_1^*$. However, the rate of energy transfer from the resonant phonons arising as a result of phonon-induced tunneling of RRCs into the thermal-phonon thermostat is slow because of the trapping of resonant phonons by two-level tunneling systems. This leads to a resulting rate of SLR of RRCs in the form

$$\frac{1}{\tau_1^*} = \frac{1}{T_1^*}\left(\frac{\dfrac{1}{\tau_{ph}}}{\dfrac{1}{\tau_{ph}} + \dfrac{1}{\tau_{pit}}}\right), \tag{24}$$

where the second cofactor is related with the trapping of resonant phonons and has a very weak temperature dependence. Indeed, $1/\tau_{pit}$ of resonant phonons does not depend on the temperature, and in Eq. (23) for $1/\tau_{ph}$ only $1/\tau_{ph}^{(a)} \sim T$ for an actual third-order anharmonic process is temperature-dependent. The scheme of SLR of RRCs is displayed in Fig. 6. We note that the expression (24) reduces to the expression (4) when $1/\tau_{ph} \gg 1/\tau_{pit}$, i.e., when thermodynamic equilibrium is established sufficiently rapidly in the phonon system.

It follows directly from Eqs. (23) and (24) that for sufficiently large samples, where $v_l < 1/\tau_{ph}^{(a)}$, the size-dependence $1/\tau_1^*(l)$ can be neglected. However, at the critical value $l = l_{cr}$, where

$$l_{cr} = v\,\tau_{ph}^{(a)}, \tag{25}$$

as l increases, $1/\tau_1^*$ increases rapidly right up to $1/T_1^*$. In our case it must be assumed that $l_{cr} \approx 0.5$ mm. This gives a reasonable value $1/\tau_{ph}^{(a)} \approx 0.46 \cdot 10^7$ Hz at $T = 1.6$ K. Therefore, the experimentally observed size-dependence of the SLR rate can be associated to the trapping of resonant phonons by two-level tunneling systems, which competes with the process of thermalization of the thermal phonons.

In conclusion, we note that although for $l < l_{cr}$ the mechanism of phonon scattering by the boundaries of the sample becomes important, the phonons are also scattered as they propagate toward the walls. Indeed, in the case of the ballistic regime the phonon lifetime T_{ph}, determining the SLR time under the phonon bottleneck conditions, would be $T_{ph} \approx l_{cr}/v \approx 2 \cdot 10^{-7}$ s^{-1}. Experimentally, however, $T_{ph} \sim T_1 \approx 10^{-3}$ s. The maximum size of a nonuniformity can be estimated if the phonons are assumed to be scattered by a Rayleigh mechanism. For the Rayleigh mechanism the phonon wavelength λ_{ph} must be much greater than the size δ of a defect. In our case $\lambda_{ph} \approx v/\nu \approx 300$ nm ($\nu \approx 10^{10}$ s^{-1}), i.e., $\delta \ll 300$ nm. Our experiments with heat pulses[25,26] also confirm the fact that in this glass the defects are much smaller than 300 nm. In these experiments the diffusion propagation of phonons with energy $E \sim (7\text{–}50)$ cm^{-1} at a rate of ~ 1.0 m/s in Er^{3+}-doped phosphate glass was recorded by the EPR method. We determined the phonon energy $E \sim (7\text{–}50)$ cm^{-1} on the basis of optical measurements of the energy and width of the lowest excited levels of Er^{3+} ions in the phosphate glass[27] as well as on the basis of estimates of the energies of the phonons in the heat pulse. A phonon with energy $E \approx 10$ cm^{-1} possesses a wavelength $\lambda_{ph} \approx 10$ nm. Therefore, the defects can be of the order of several nm in size. The sizes of the micrononuniformities in the glasses studied agree with the results of Ref. 1.

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